

# Modelling zero-inflated (and overdispersed) count data

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Francisco Rodríguez-Sánchez

<https://frodriguezsanchez.net>

# How many eggs in nests?



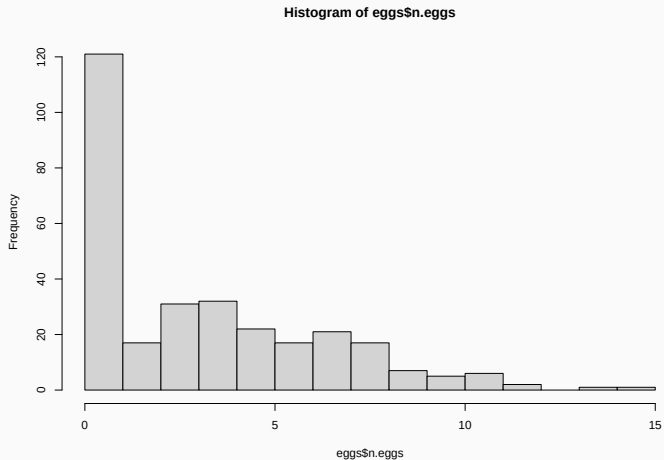
```
eggs <- read.csv('data/eggs.csv')
```

| diameter | old | n.eggs |
|----------|-----|--------|
| 14       | no  | 4      |
| 8        | yes | 0      |
| 7        | yes | 0      |

**diameter:** nest diameter (cm)

**old:** does nest look old/abandoned?

## How many eggs in nests?



Many zeros does not mean you need a zero-inflated model!

Check model afterwards

## How many eggs in nests?

- Nests may be occupied or not

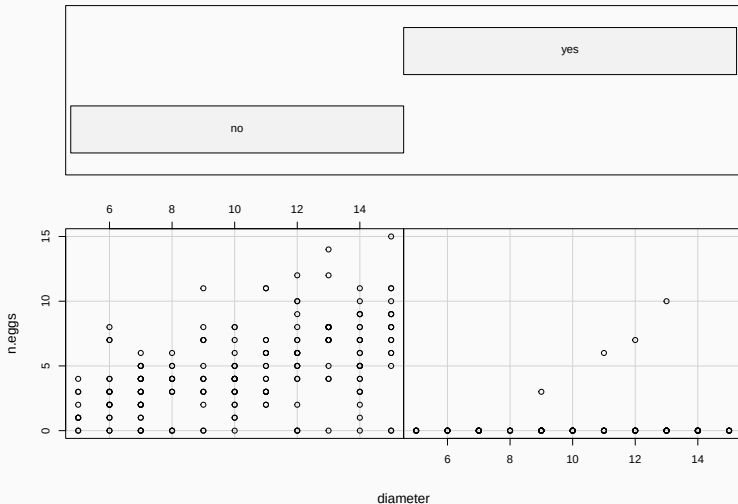
## How many eggs in nests?

- Nests may be occupied or not
- Occupied nests may not have eggs (too soon, predation, etc)

Number of eggs ~ nest diameter \* old appearance

```
coplot(n.eggs ~ diameter | old, data = eggs)
```

Given : old



```
eggs.poi <- glm(n.eggs ~ old * diameter,  
               data = eggs,  
               family = poisson)
```

# Trying Poisson GLM

Call:

```
glm(formula = n.eggs ~ old * diameter, family = poisson, data = eggs)
```

Coefficients:

|                 | Estimate | Std. Error | z value | Pr(> z )     |
|-----------------|----------|------------|---------|--------------|
| (Intercept)     | 0.30773  | 0.12883    | 2.389   | 0.0169 *     |
| oldyes          | -3.78879 | 0.92230    | -4.108  | 3.99e-05 *** |
| diameter        | 0.11441  | 0.01105    | 10.354  | < 2e-16 ***  |
| oldyes:diameter | 0.08513  | 0.07634    | 1.115   | 0.2648       |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

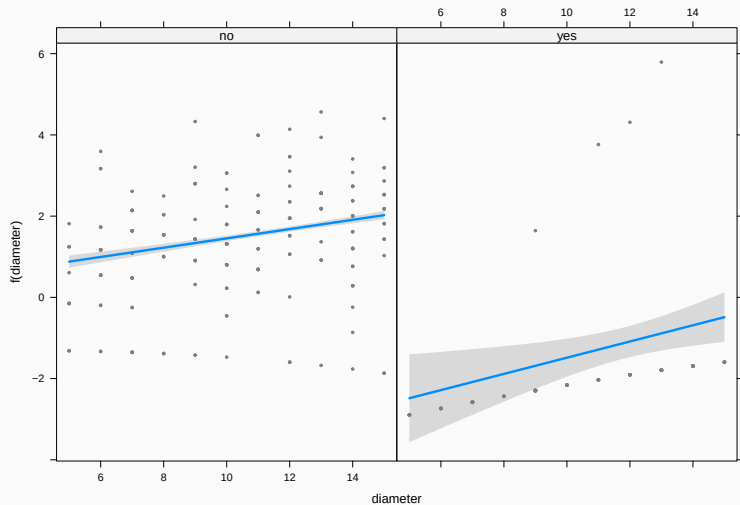
(Dispersion parameter for poisson family taken to be 1)

Null deviance: 1184.57 on 299 degrees of freedom  
Residual deviance: 526.97 on 296 degrees of freedom  
AIC: 1176.7

Number of Fisher Scoring iterations: 7



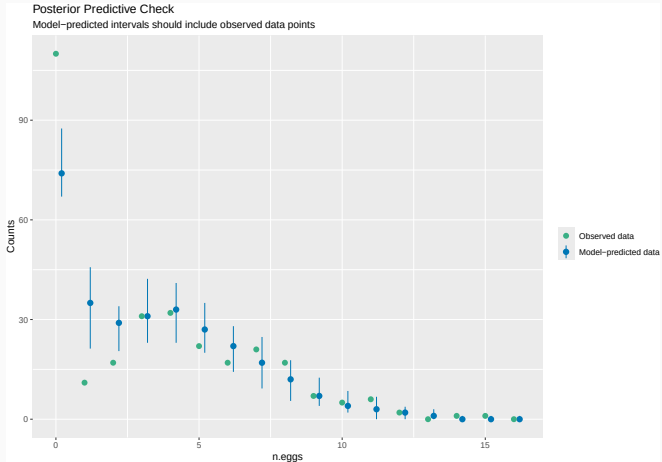
# Visualising the fitted Poisson GLM



# Checking Poisson GLM

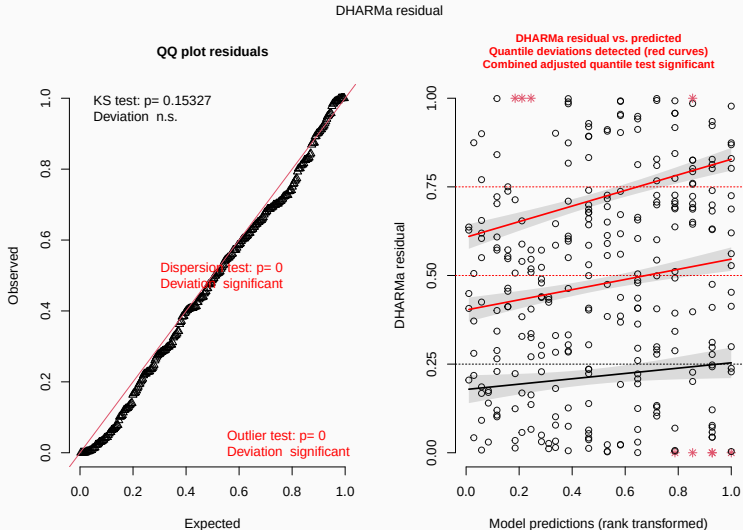
Simulate data from fitted model (**yrep**) and compare with observed data (**y**)

```
library('easystats')  
check_predictions(eggs.poi)
```

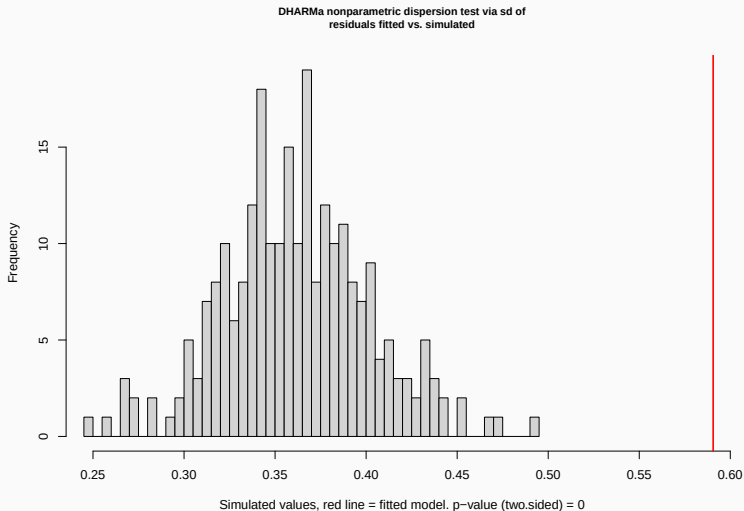


# Checking Poisson GLM with DHARMA

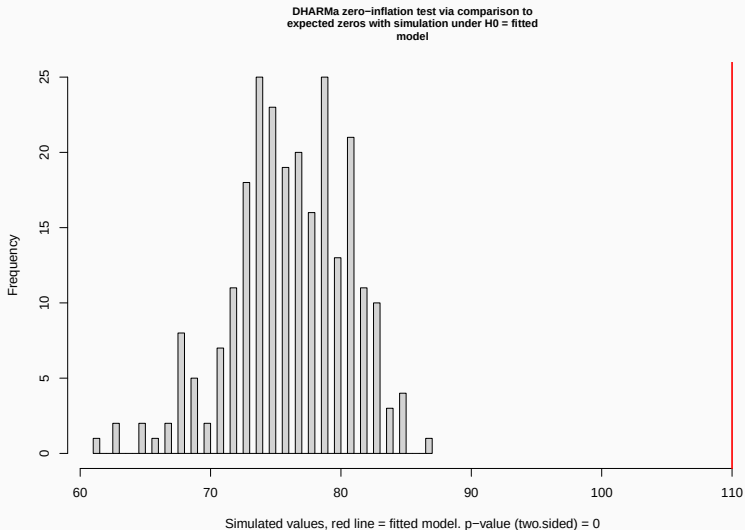
```
library('DHARMA')  
eggs.poi.res <- simulateResiduals(eggs.poi, plot = TRUE)
```



```
testDispersion(eggs.poi.res)
```



```
testZeroInflation(eggs.poi.res)
```



## Accounting for zero-inflation

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Mixture model:

1. Model probability of 0 (Binomial)

Mixture model:

1. Model probability of 0 (Binomial)
2. Model counts (including 0) (Poisson/Negative Binomial)



# Modelling egg number as Zero-Inflated Poisson (ZIP)

Nests may be occupied or not:

*Probability nest not occupied ~ old* (Binomial)

For occupied nests:

*Number of eggs ~ Nest diameter* (Poisson)

```
library('glmmTMB')
eggs.zip <- glmmTMB(n.eggs ~ diameter,
                    family = 'poisson',
                    ziformula = ~ old,
                    data = eggs)
```

# Modelling egg number as Zero-Inflated Poisson

```
Family: poisson ( log )
Formula:          n.eggs ~ diameter
Zero inflation:    ~old
Data: eggs
```

| AIC   | BIC    | logLik | deviance | df.resid |
|-------|--------|--------|----------|----------|
| 993.8 | 1008.6 | -492.9 | 985.8    | 296      |

Conditional model:

|             | Estimate | Std. Error | z value | Pr(> z )    |
|-------------|----------|------------|---------|-------------|
| (Intercept) | 0.41622  | 0.13619    | 3.056   | 0.00224 **  |
| diameter    | 0.11248  | 0.01155    | 9.737   | < 2e-16 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

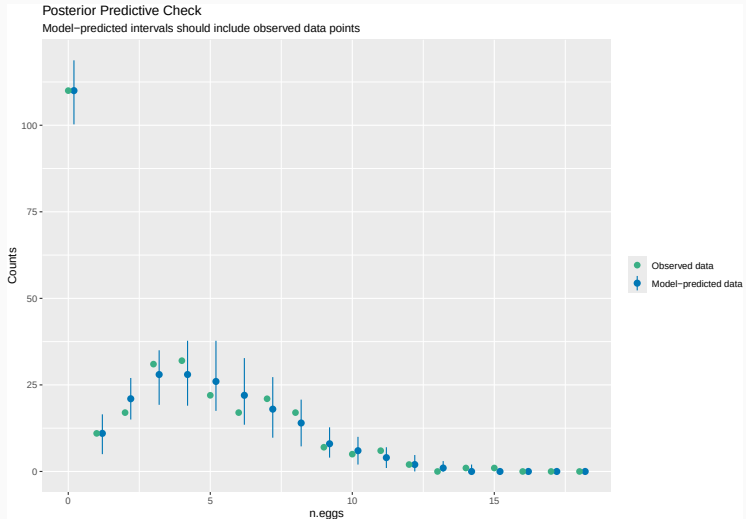
|             | Estimate | Std. Error | z value | Pr(> z )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | -2.4054  | 0.2803     | -8.582  | <2e-16 *** |
| oldyes      | 5.4897   | 0.5830     | 9.416   | <2e-16 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

# Checking ZIP model

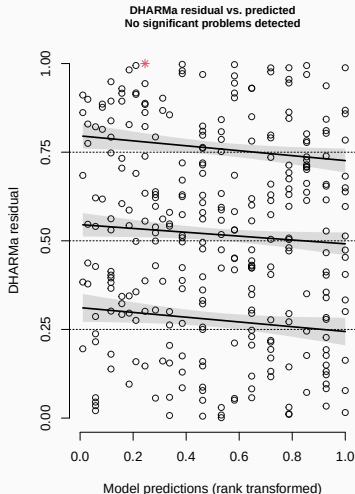
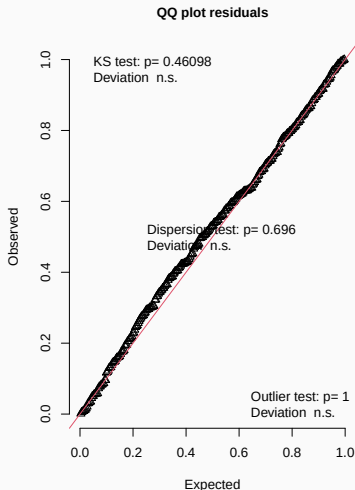
```
check_predictions(eggs.zip)
```



# Checking ZIP model with DHARMA

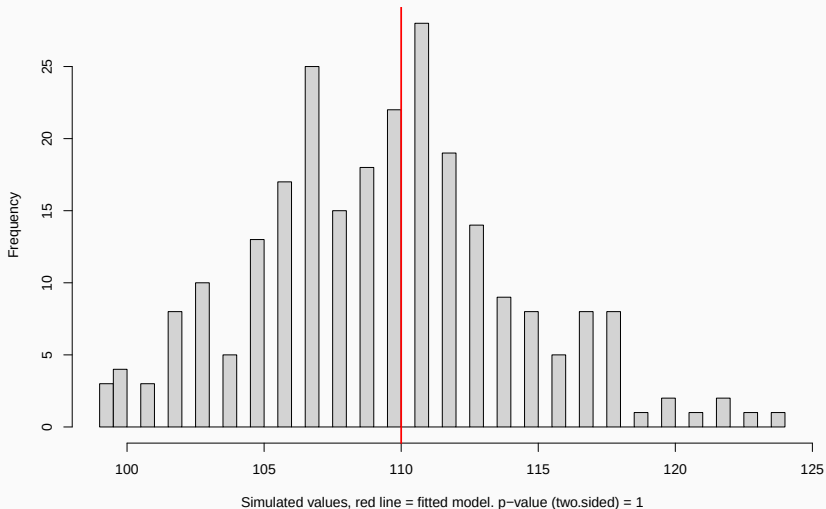
```
eggs.zip.res <- simulateResiduals(eggs.zip, plot = TRUE)
```

DHARMA residual

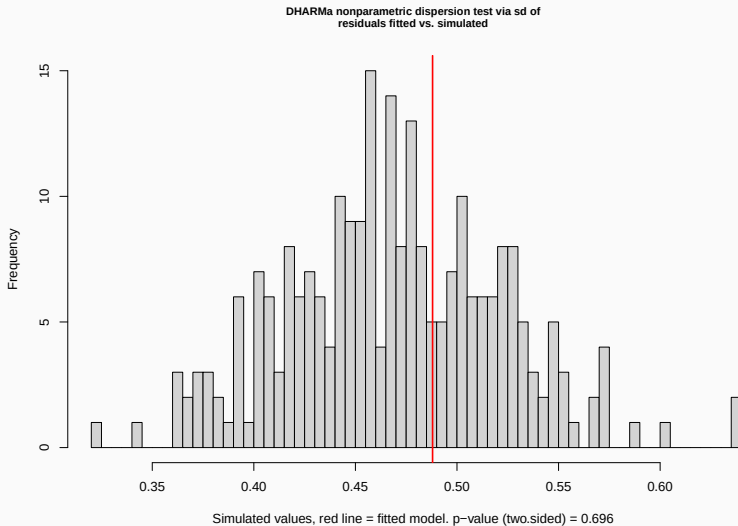


```
testZeroInflation(eggs.zip.res)
```

DHARMA zero-inflation test via comparison to  
expected zeros with simulation under  $H_0$  = fitted  
model



```
testDispersion(eggs.zip.res)
```



## Modelling egg number as Zero-Inflated Negative Binomial (ZINB)

(If there were overdispersion with Poisson)

```
eggs.zinb <- glmmTMB(n.eggs ~ diameter,  
                     family = 'nbinom2',  
                     ziformula = ~ old,  
                     data = eggs)
```

# Modelling egg number as ZINB

```
Family: nbinom2 ( log )
Formula:          n.eggs ~ diameter
Zero inflation:    ~old
Data: eggs
```

| AIC   | BIC    | logLik | deviance | df.resid |
|-------|--------|--------|----------|----------|
| 995.7 | 1014.2 | -492.8 | 985.7    | 295      |

Dispersion parameter for nbinom2 family (): 143

Conditional model:

|             | Estimate | Std. Error | z value | Pr(> z )    |
|-------------|----------|------------|---------|-------------|
| (Intercept) | 0.4118   | 0.1389     | 2.964   | 0.00304 **  |
| diameter    | 0.1128   | 0.0118     | 9.561   | < 2e-16 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

|             | Estimate | Std. Error | z value | Pr(> z )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | -2.4160  | 0.2846     | -8.489  | <2e-16 *** |
| oldyes      | 5.4995   | 0.5850     | 9.401   | <2e-16 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1



# Comparing models

```
library('parameters')  
compare_models(eggs.poi, eggs.zip, eggs.zinb)
```

| Parameter            |  | eggs.poi             |  | eggs.zip          |  | eggs.zinb         |
|----------------------|--|----------------------|--|-------------------|--|-------------------|
| (Intercept)          |  | 0.31 ( 0.06, 0.56)   |  | 0.42 (0.15, 0.68) |  | 0.41 (0.14, 0.68) |
| diameter             |  | 0.11 ( 0.09, 0.14)   |  | 0.11 (0.09, 0.14) |  | 0.11 (0.09, 0.14) |
| old [yes]            |  | -3.79 (-5.60, -1.98) |  |                   |  |                   |
| old [yes] × diameter |  | 0.09 (-0.06, 0.23)   |  |                   |  |                   |
| Observations         |  | 300                  |  | 300               |  | 300               |

# Comparing models

```
library('performance')
compare_performance(eggs.poi, eggs.zip, eggs.zinb)
```

```
# Comparison of Model Performance Indices
```

| Name      | Model   | AIC (weights)  | AICc (weights) | BIC (weights)  | RMSE  |
|-----------|---------|----------------|----------------|----------------|-------|
| eggs.poi  | glm     | 1176.7 (<.001) | 1176.8 (<.001) | 1191.5 (<.001) | 2.324 |
| eggs.zip  | glmmTMB | 993.8 (0.719)  | 993.9 (0.726)  | 1008.6 (0.942) | 2.324 |
| eggs.zinb | glmmTMB | 995.7 (0.281)  | 995.9 (0.274)  | 1014.2 (0.058) | 2.324 |

| Name      | Sigma   | R2    | R2 (adj.) | Score_log | Score_spherical | Nagelkerke' |
|-----------|---------|-------|-----------|-----------|-----------------|-------------|
| eggs.poi  | 1.000   |       |           | -1.948    | 0.042           | 0           |
| eggs.zip  | 1.000   | 0.118 | 0.112     | -1.643    | 0.040           |             |
| eggs.zinb | 143.279 | 0.118 | 0.112     |           |                 |             |

## Accounting for zero-inflation with hurdle models

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# Tracking measles outbreak

Counting number of hives/person

Many people not sick (0 hives)

Those sick, have many hives ( $>1$ )



ZIP/ZINB:

1. Binomial model: probability of zero

Hurdle:

### ZIP/ZINB:

1. Binomial model: probability of zero
2. Count model (Poisson/NegBin) includes zero

### Hurdle:

### ZIP/ZINB:

1. Binomial model: probability of zero
2. Count model (Poisson/NegBin) includes zero

### Hurdle:

1. Binomial model: probability of non-zero

### ZIP/ZINB:

1. Binomial model: probability of zero
2. Count model (Poisson/NegBin) includes zero

### Hurdle:

1. Binomial model: probability of non-zero
2. Count model truncated at 1

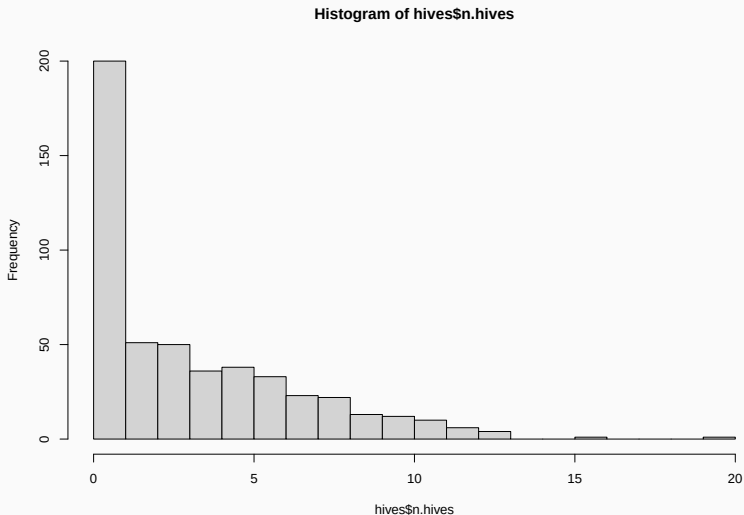


## How many hives per skin area?

```
hives <- read.csv('data/hives.csv')
```

| age          | vaccinated    | area.cm2       | n.hives        |
|--------------|---------------|----------------|----------------|
| Min. : 1.0   | Min. :0.000   | Min. : 5.000   | Min. : 0.000   |
| 1st Qu.:23.0 | 1st Qu.:0.000 | 1st Qu.: 6.000 | 1st Qu.: 0.000 |
| Median :45.0 | Median :1.000 | Median : 8.000 | Median : 2.000 |
| Mean :44.7   | Mean :0.648   | Mean : 7.482   | Mean : 3.256   |
| 3rd Qu.:65.0 | 3rd Qu.:1.000 | 3rd Qu.: 9.000 | 3rd Qu.: 5.250 |
| Max. :90.0   | Max. :1.000   | Max. :10.000   | Max. :20.000   |

## Many people with 0 hives

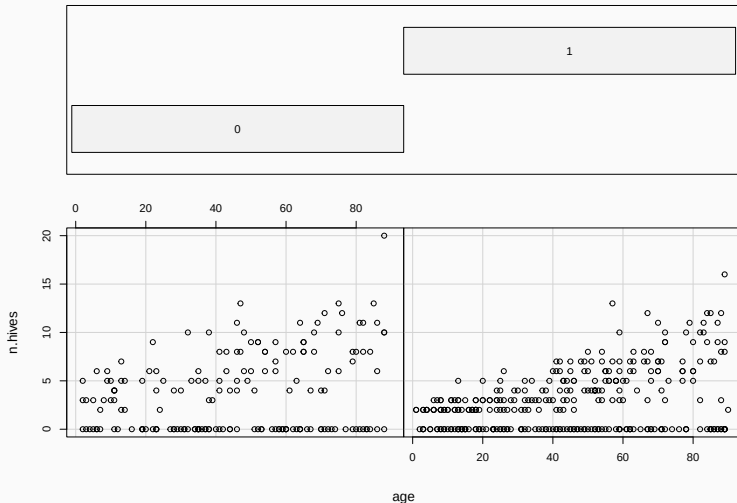


(that does not mean we need zero-inflated model!)

# Number of hives ~ age \* vaccinated

```
coplot(n.hives ~ age | as.factor(vaccinated), data = hives)
```

Given : as.factor(vaccinated)



```
hives.poi <- glm(n.hives ~ vaccinated * age,  
                 offset = log(area.cm2),  
                 data = hives,  
                 family = poisson)
```

# Trying Poisson GLM

Call:

```
glm(formula = n.hives ~ vaccinated * age, family = poisson, data = hives,  
     offset = log(area.cm2))
```

Coefficients:

|                | Estimate  | Std. Error | z value | Pr(> z )    |
|----------------|-----------|------------|---------|-------------|
| (Intercept)    | -1.363696 | 0.095097   | -14.340 | < 2e-16 *** |
| vaccinated     | -0.334184 | 0.122887   | -2.719  | 0.00654 **  |
| age            | 0.013626  | 0.001623   | 8.395   | < 2e-16 *** |
| vaccinated:age | 0.002034  | 0.002075   | 0.980   | 0.32708     |

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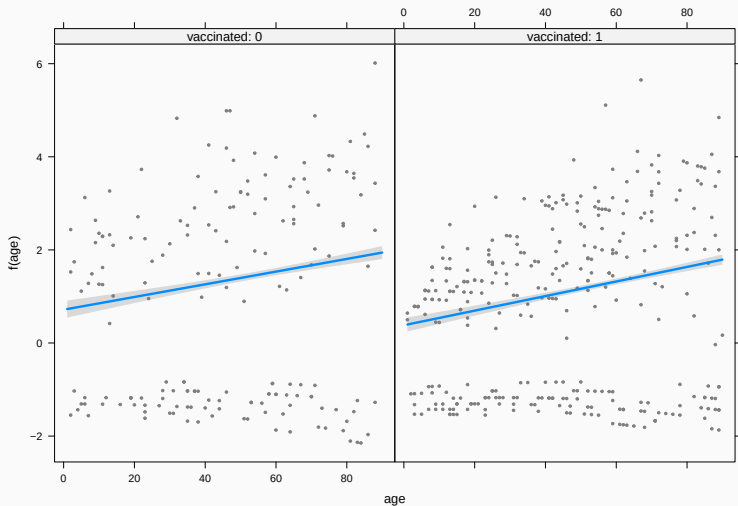
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 2137.0 on 499 degrees of freedom  
Residual deviance: 1891.7 on 496 degrees of freedom  
AIC: 2925.6

Number of Fisher Scoring iterations: 5

# Visualising fitted Poisson GLM

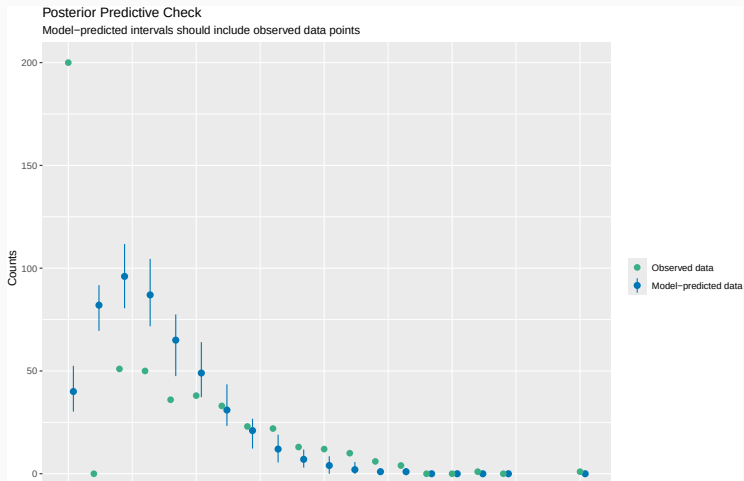


# Checking Poisson GLM

```
check_predictions(hives.poi)
```

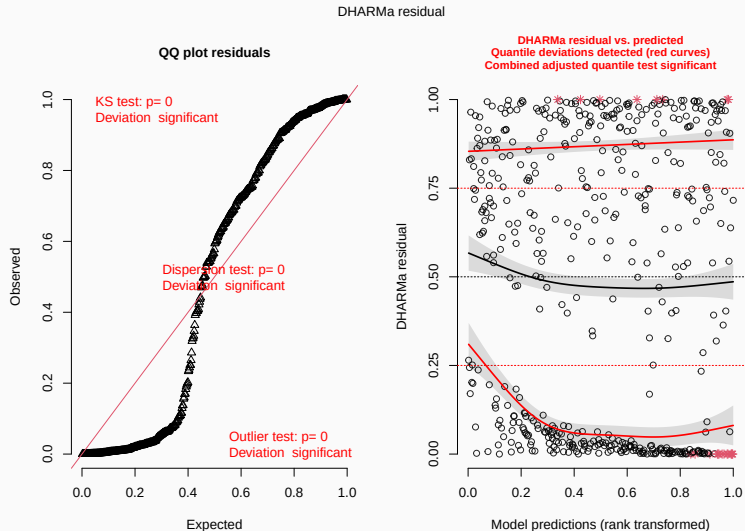
Warning: Maximum value of original data is not included in the replicated data.

Model may not capture the variation of the data.



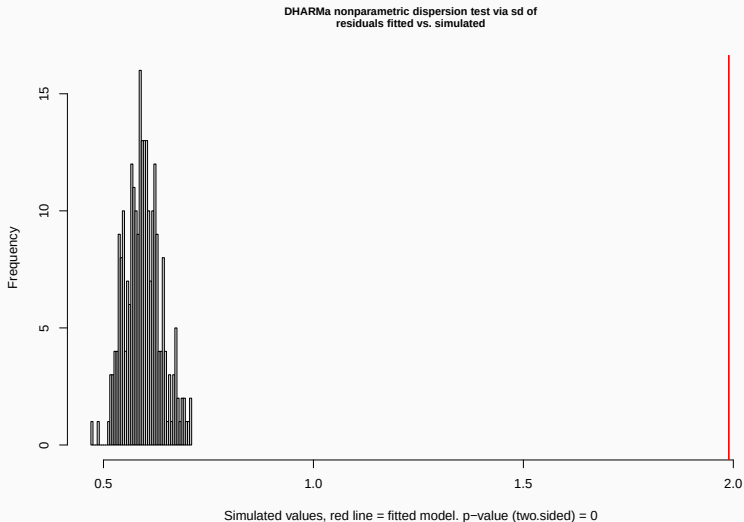
# Checking Poisson GLM

```
hives.poi.res <- simulateResiduals(hives.poi, plot = TRUE)
```



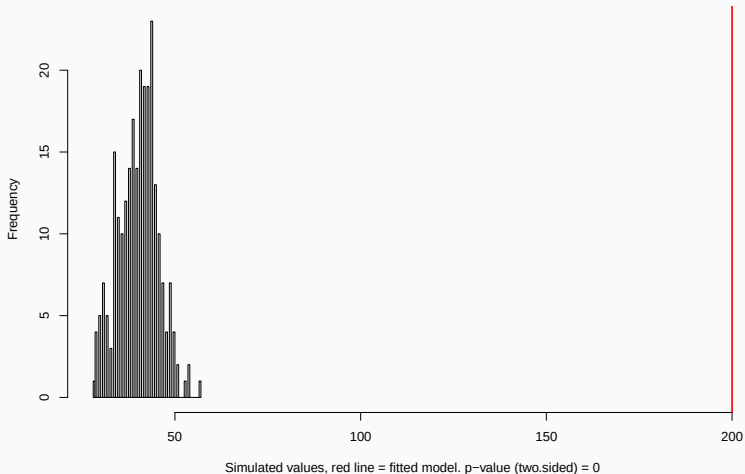


```
testDispersion(hives.poi.res)
```



```
testZeroInflation(hives.poi.res)
```

DHARMA zero-inflation test via comparison to  
expected zeros with simulation under  $H_0$  = fitted  
model



## Accounting for zero-inflation with hurdle model

```
hives.hur <- glmmTMB(n.hives ~ vaccinated + age,  
                     family = truncated_poisson,  
                     ziformula = ~ 1,  
                     offset = log(area.cm2),  
                     data = hives)
```

# Accounting for zero-inflation with hurdle model

```
Family: truncated_poisson ( log )
Formula:          n.hives ~ vaccinated + age
Zero inflation:    ~1
Data: hives
Offset: log(area.cm2)
```

| AIC    | BIC    | logLik | deviance | df.resid |
|--------|--------|--------|----------|----------|
| 1932.1 | 1949.0 | -962.1 | 1924.1   | 496      |

Conditional model:

|             | Estimate  | Std. Error | z value | Pr(> z )     |
|-------------|-----------|------------|---------|--------------|
| (Intercept) | -0.853885 | 0.070755   | -12.068 | < 2e-16 ***  |
| vaccinated  | -0.365664 | 0.051532   | -7.096  | 1.29e-12 *** |
| age         | 0.014860  | 0.001065   | 13.955  | < 2e-16 ***  |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

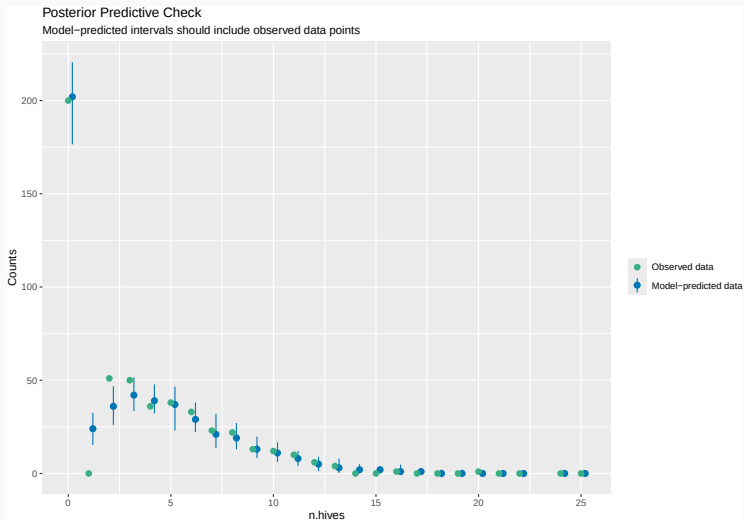
|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -0.40547 | 0.09129    | -4.442  | 8.93e-06 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

# Checking hurdle model

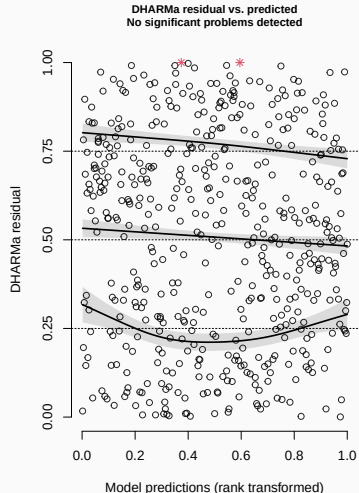
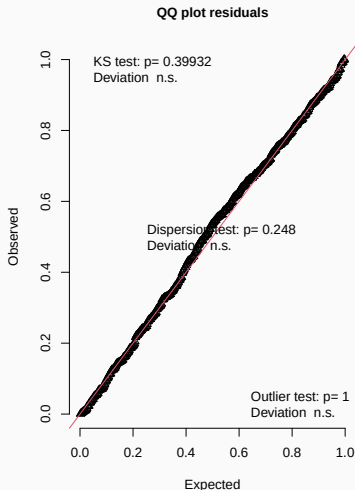
```
check_predictions(hives.hur)
```



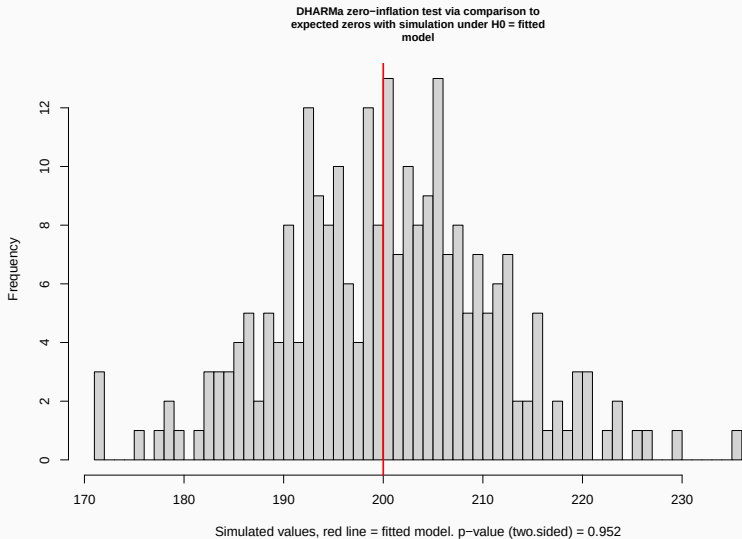
# Checking hurdle model with DHARMA

```
hives.hur.res <- simulateResiduals(hives.hur, plot = TRUE)
```

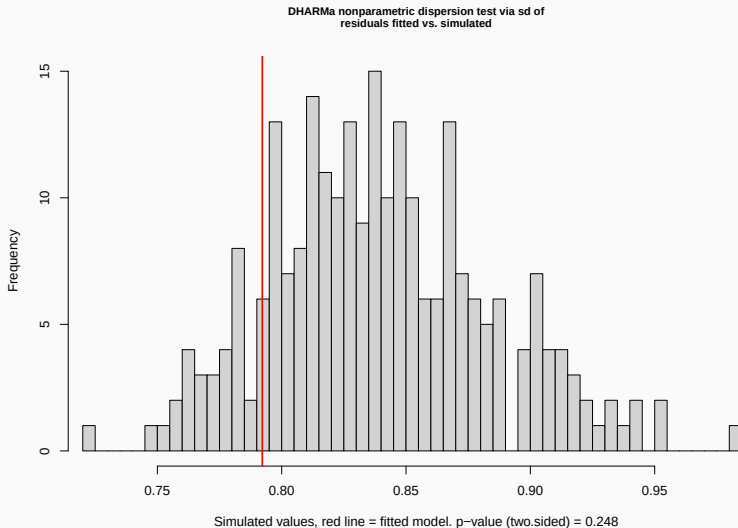
DHARMA residual



```
testZeroInflation(hives.hur.res)
```



```
testDispersion(hives.hur.res)
```





# Comparing models

```
compare_models(hives.poi, hives.hur)
```

| Parameter        | hives.poi              | hives.hur            |
|------------------|------------------------|----------------------|
| (Intercept)      | -1.36 (-1.55, -1.18)   | -0.85 (-0.99, -0.72) |
| vaccinated       | -0.33 (-0.58, -0.09)   | -0.37 (-0.47, -0.26) |
| age              | 0.01 ( 0.01, 0.02)     | 0.01 ( 0.01, 0.02)   |
| vaccinated × age | 2.03e-03 ( 0.00, 0.01) |                      |
| Observations     | 500                    | 500                  |

# Comparing models

```
compare_performance(hives.poi, hives.hur)
```

```
# Comparison of Model Performance Indices
```

| Name      | Model   | AIC (weights)  | AICc (weights) | BIC (weights)  | RMSE  |
|-----------|---------|----------------|----------------|----------------|-------|
| hives.poi | glm     | 2925.6 (<.001) | 2925.7 (<.001) | 2942.5 (<.001) | 3.299 |
| hives.hur | glmmTMB | 1932.1 (>.999) | 1932.2 (>.999) | 1949.0 (>.999) | 3.310 |

| Name      | Sigma | Score_log | Score_spherical | Nagelkerke's R2 | R2    | R2 (adj) |
|-----------|-------|-----------|-----------------|-----------------|-------|----------|
| hives.poi | 1.000 | -2.918    | 0.034           | 0.393           |       |          |
| hives.hur | 1.000 | -2.246    | 0.035           |                 | 0.498 | 0.4      |